FNP Data Analytics: Project Steps

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Project overview

I built an interactive Excel dashboard on a synthetic FNP (Ferns N Petals) dataset of 1,000 gift orders from 2023, to find what drives revenue and how quickly orders are delivered.

Business questions

1. Which categories and occasions bring in the most revenue?
2. How does revenue change month to month?
3. Which products and cities perform best?
4. Which products are most popular for each occasion?
5. How long do deliveries take, and does order size affect delivery time?

Dataset

File	Rows	Contents
orders.csv	1,000	Order and delivery dates and times, quantity, city, occasion
customers.csv	100	Customer name, city, contact, gender, address
products.csv	70	Product name, category, price, occasion

Tools: Excel Power Query, Data Model (Power Pivot) with DAX, PivotTables and PivotCharts, slicers, a timeline and the CORREL function.

Step 1: Import the CSV files with Power Query

I connected Excel to the project's Data folder once, so all three files load and refresh from one place.

1. **Data → Get Data → From File → From Folder**, pointing at the folder that holds the three CSVs. I loaded this folder listing to the **Data** sheet (file name, extension, dates, folder path).

2. From the folder query, I created one query per file by opening its **Content**: **Customers**, **Orders** and **Products**.
3. In each query, Power Query read the CSV (comma-delimited) and **promoted the first row to headers**.
4. I loaded Customers, Orders and Products to their own sheets, and added Orders and Customers to the **Data Model** (Step 4).

Step 2: Clean and shape the data in Power Query

I set the right data type on every column and brought product details into the Orders table, so each order row holds everything the analysis needs.

Query	Step	What it does
All three	Changed Type	IDs, quantity and price as whole numbers; dates as Date; times as Time; names and text as Text
Products	Removed Columns	Dropped **Description**, which no analysis uses
Orders	Merge Queries	Left outer join to Products on **Product_ID**
Orders	Expand	Kept **Product_Name**, **Category** and **Price (INR)** from Products

The Product_ID merge means every order now carries its product's name, category and price, which Step 3 needs to calculate revenue.

Step 3: Add calculated columns

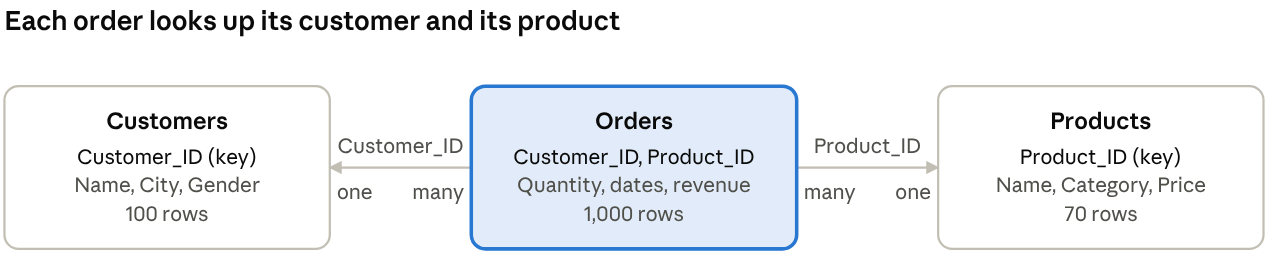
I added five columns to the Orders query in Power Query (**Add Column** tab), so they recalculate on every refresh.

Column	How it's calculated	Used for
Month Name	Date → Month → Name of Month, from Order_Date	Revenue per month
Day Name	Date → Day → Name of Day, from Order_Date	Day-of-week analysis

Column	How it's calculated	Used for
Hour	Time → Hour, from Order_Time	Order-time analysis
Days Required for Delivery	Custom column `[Delivery_Date] - [Order_Date]`, converted to total days, whole number	Average delivery days, correlation
Total_Order_Revenue	Custom column `[Quantity] * [Price (INR)]`, Currency type	All revenue figures

Step 4: Build the Data Model and a DAX measure

I loaded the three tables into the Data Model and related them, so one pivot can combine order, product and customer fields.



Data Model · 3 tables, 2 relationships

Orders sits in the middle; each order row points to one customer and one product, so a pivot can mix fields from all three tables.

1. Added **Orders**, **Customers** and **Products** to the Data Model (Load To → Add this data to the Data Model).
2. Created two relationships in **Power Pivot → Diagram View**:
   a. Orders[Customer_ID] → Customers[Customer_ID] (many orders to one customer)
   b. Orders[Product_ID] → Products[Product_ID] (many orders to one product)
3. Added a measure to show revenue in lakh. Calculated Field is unavailable for Data Model pivots, so I used **Add Measure** on the Orders table:

```
Revenue (L) = SUM(Orders[Total_Order_Revenue]) / 100000
```

4. Formatted the measure's cells with the custom number format `"₹"0.00" L"`, so ₹35,20,984 shows as **₹35.21 L**.

Step 5: Analyse with PivotTables

I built eight PivotTables from the Data Model on the **Calculations** sheet, one per business question.

Pivot	Rows	Values	Filter / sort	Answers
KPIs	—	Revenue (L), Average of Days Required for Delivery, Average of Total_Order_Revenue	—	Total revenue, delivery days, average order value
Revenue by category	Category	Revenue (L)	—	Which categories earn most
Revenue by occasion	Occasion	Revenue (L)	Sorted by revenue	Which occasions earn most
Revenue by month	Month Name	Revenue (L)	Calendar order	Monthly trend
Top 10 items	Product_Name	Revenue (L)	Top 10, sorted largest first	Best-selling products
Top 10 cities	Customer City	Revenue (L)	Top 10, sorted largest first	Best cities
Product popularity	Occasion → Product_Name	Sum of Quantity	Top 3 per occasion, sorted largest first	Most popular products per occasion
Revenue by gender	Customer Gender	Sum of Total_Order_Revenue	—	Revenue split by gender

How I set the filters and sorting

- **Top N:** Row Labels ▼ → Value Filters → Top 10, then set the count (10, or 3 for products per occasion). With Occasion above Product_Name, the Top 3 applies within each occasion.
- **Sort:** clicked a single value cell → Sort Largest to Smallest. Sorting is greyed out when several cells are selected.

Step 6: Build the dashboard

I built the **Dashboard** sheet from three KPI cards, six PivotCharts, a slicer and a timeline, all fed by the Step 5 pivots.

KPI cards: rounded shapes linked to the KPI pivot cells, so they update with the filters. I selected each shape and typed `=` and the cell in the formula bar.

Card	Linked cell	Shows
Total Revenue	Calculations!I3	₹35.21 L
Average Delivery Days	Calculations!J3	5.53
Average Order Value	Calculations!K3	₹3,521

Charts: each built with PivotTable Analyze → PivotChart, then moved to the Dashboard sheet.

Chart	Source pivot	Chart type
Revenue by Category	Revenue by category	3-D column
Occasions vs Revenue	Revenue by occasion	3-D bar, categories in reverse order so the largest is on top
Revenue per Month	Revenue by month	Line
Top 10 Items	Top 10 items	3-D column
Top 10 Cities	Top 10 cities	Column
Product Popularity by Occasions	Product popularity	Column, grouped by occasion

Filters

- **Occasion slicer** (Insert Slicer → Occasion), connected through Report Connections to the KPI, category, month, top items, top cities and gender pivots. The occasions chart is left unconnected so it always compares all seven occasions.
- **Order_Date timeline** (Insert Timeline), connected to every pivot except product popularity.

Formatting: one navy colour across cards and charts, data labels in lakh, bars sorted by value, and a **Key Insights** text block under the KPI cards.

Step 7: Check the data and write the insights

I tested the delivery question with a correlation, checked the data for traps that could mislead a chart, then wrote the findings on the dashboard and in an Executive Summary.

Correlation test: quantity vs delivery days in the Orders sheet.

```
=CORREL(Orders!D2:D1001, Orders!N2:N1001)
```

The result, **r = 0.0035**, means order size has no effect on delivery time. Average delivery stays between 5.25 and 5.83 days for every quantity from 1 to 5.

Data checks

Check	What I found	What I did
Duplicate product names	Quia Gift (IDs 21, 56) and Error Gift (IDs 65, 70)	Treated as one product each: a data-entry typo, per the data instructions
Product vs occasion	Every product is sold for one occasion only	Ranked product popularity within each occasion
"All Occasions" in the slicer	It's a real occasion value (126 orders), not "select all"	Cleared the slicer filter to show all data

Insights

1. Colors leads with 29% of revenue (₹10.06 L); Soft Toys and Sweets follow at about 21% each.
2. February (₹7.05 L) and August (₹7.37 L) are peak months, around Valentine's Day and Raksha Bandhan.
3. Anniversary (₹6.75 L) and Raksha Bandhan (₹6.32 L) are the biggest occasions.
4. Magnam Set (₹1.22 L) is the top product; Imphal (₹1.26 L) is the top city.
5. Order size doesn't affect delivery time (r = 0.00).

Insights 1, 2 and 5 went into the dashboard's **Key Insights** block. The **Executive Summary** sheet and the insights PDF hold the full list with recommendations, drafted with AI assistance and checked against the data.

Step 8: Publish the project on GitHub

I published the project at [sathvik200/FNP_Data_Analytics] so recruiters can open the files and see the dashboard.

File	Contents
FNP_Data_Analytics.xlsx	The workbook: queries, Data Model, pivots, dashboard
Dashboard.png	Screenshot of the finished dashboard
AI Assisted Insights.pdf	Insights and recommendations
Data/	The three raw CSV files